

ARITHMETIC AND CALIBRATION

CI head f028ec5 / merge ae11d8b | Verilator 5.020 | Group 1/3

21 RTL benches passed; independent arithmetic audit passed

RTL CI job succeeded; workflow conclusion is separate

BENCH / OBSERVED LOG COUNTS	REFERENCE CHECK / COVERAGE BOUNDARY
p2_oracle_tb 1,048,576 input codes; 0 errors Observed: P2_ORACLE_COMPLETE codes=1048576 errors=0	Identity: dout == input code; zero flags and 11-cycle completion. One-slice fixture; not all configurations.
divider_borrow_tb 67,592 operand/config combinations 6 width/unroll fixtures Observed: DIVIDER_BORROW_COMPLETE exhaustive_checks=67592	Signed floor reference + defining inequality; zero divisor, busy and hold. Six small-width parameter fixtures.
cal_weight_reduce_ppa_tb 1,430,848 miter input cases / 7 fixtures 43,758 allocations in 18-slice fixture Observed: CAL_WEIGHT_REDUCE_PPA_COMPLETE	Independent physical-cell scalar sum; old/new T/G/R and invalid-ID checks. Transaction-edge sampling; no mapped/STA claim.
calibration_physical_tb 2,048 transactions; 18 physical slices Observed: CALIBRATION_PHYSICAL_COMPLETE	Independent 256-bit integer formula; code, flags, ID and input capture. 18 slices, reduced 8-cell test geometry.
calibration_recovery_tb 2,048 target/residue transactions max error : calibrated 4; nominal 504 codes Observed: CALIBRATION_RECOVERY_COMPLETE	Known target -> synthetic residue -> calibrated and nominal-weight outputs. Synthetic mismatch + ideal ADC2 quantization.
calibration_fit_tb 128 holdout samples; 1,278 weights code + flags + ID + 11-cycle latency Observed: CALIBRATION_FIT_COMPLETE	Frozen external holdout vectors; code/flags/ID/latency + coefficient readback. Fitted coefficients loaded, not learned in RTL.
p2_tb 23,049 TB checks; 0 errors Observed: P2 RESULT: PASS	Golden vectors + floor inequality; rounding, masks, carry, clipping and link. T5 full-code sweep is a separate bench.

Counts retain their own units. No combined 'total coverage' score. All 21 log and TB hashes verified.

Source: docs/evidence/20260930/ci_rtl_f028/ | RTL only; not STA or analog performance.